

Patterns of Tobacco Specialty Store Locations and Foot Traffic in the United States

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June 2025

Abstract

Importance:

Tobacco specialty stores, including vape shops, smoke shops, cigar shops, and hookah lounges, play a crucial role in the tobacco retail landscape. However, the geographic distribution and consumer traffic patterns at these retailers remain poorly understood due to the lack of a national database tracking their locations and sales.

Objective:

To analyze access to tobacco specialty stores across the United States and examine how retailer locations and consumer visit patterns relate to socioeconomic and demographic characteristics.

Design:

Cross-sectional study combining Google Maps business listings and anonymized cell phone foot traffic data from Advan in May 2024.

Participants:

22,484 tobacco specialty stores (vape, smoke, cigar, hookah) across US census tracts categorized into vape shops (6,295), tobacco shops (13,021), cigar shops (1,507), and hookah lounges (1,661).

Main Outcomes and Methods:

Store density and monthly foot traffic at the US Census tract level.

Results:

Foot traffic was disproportionately concentrated in high-poverty and high-youth tracts. Vape stores showed the strongest association with youth demographics. One additional store is associated with a traffic volume increase of 46–117% (all $p < 0.001$), depending on store type.

Conclusion and Relevance:

Our study provides a comprehensive overview of tobacco specialty stores in the United States, and shows that poverty remains the most consistent predictor for where tobacco specialty stores locate, regardless of type.

Introduction

Tobacco specialty stores are a little understood aspect of the tobacco retail supply side. In this manuscript, we identify and analyze four important categories of tobacco specialty retailers: vape shops, smoke shops, cigar shops, and hookah lounges. These retailers may have different customer bases and use location decisions (Dai et al. 2017; Berg et al. 2020) and innovative marketing strategies to appeal to customers (Lee et al. 2018; Huang et al. 2019; Struik et al. 2020).

There is no publicly available national registry of all tobacco retailers. Tobacco specialty stores are not tracked by popular point of sale scanner systems like Nielsen and Information Resources Incorporated (IRI); as such, it is difficult to determine their relative sizes in the tobacco market as compared to larger retailers such as grocery stores and gas or convenience store chains. Liber et al. (2024) estimates that traditional sales systems miss only 10% of cigarette sales but about 50% of electronic nicotine delivery systems (ENDS) sales, thus highlighting the role these specialty retailers could play (along with online retailers). We contribute to literature identifying tobacco retailer locations by combining Google Maps data with Advan foot traffic patterns to simultaneously catalog and store locations not available in point of sale scanner data and analyze consumer foot traffic patterns among the different store types.

Due to public health concerns, access to products is a commonly discussed topic in the tobacco literature. Neighborhoods with higher concentrations of tobacco retailers tend to also have higher tobacco product consumption and lower rates of quit attempts, underscoring the importance of access (Centers for Disease Control and Prevention 2021). Policies such as Minimum Legal Sales Age (MLSA) laws, including Tobacco-21, have been shown to reduce youth access and subsequent use of tobacco products targeted by the laws (Pesko 2023;

Friedman and Pesko 2024; Abouk et al. 2024). Restrictions on access to flavored tobacco products, such as state and local bans on flavored e-cigarettes, have influenced tobacco purchasing patterns and use among young adults (Friedman et al. 2024; Brouwer et al. 2024). In some cases, policy effects may even extend beyond their immediate jurisdictions due to spatial spillovers, where consumers travel across borders to purchase restricted products (Chen et al. 2024).

Prior research has relied on the number of retailers as a measure of access. For example, counts of tobacco product retail locations including gas and convenience, grocery, liquor, pharmacy, discount stores, and supermarkets increased by 12% in the United States from 2010-2017 (Golden et al. 2022). However, this approach treats all stores as equal, overlooking substantial variation in store size, popularity, and sales volume. A single high-volume retailer may facilitate access as much as several smaller outlets. Since direct measures of store volume (e.g., square footage, inventory, or sales) are not systematically available for tobacco specialty stores, we use foot traffic as measured by cell phone mobility collected by Advan via anonymized mobile device visits as a proxy for retail exposure and potential consumer access. This measure captures not only the presence of a retailer, but also how actively consumers engage with it, offering a more granular and behaviorally relevant indicator of access.

Studies which have identified vape, tobacco, or cigar shops have also mostly focused on limited geographies such as selected counties or Metropolitan Statistical Areas (Bostean et al. 2018; Center for Public Health Systems Science 2014; Lee et al. 2016; Berg et al. 2020; Barker et al. 2021). At the national level, some previous literature has developed methods for identifying likely retailers of certain products such as premium cigars in the now decommissioned NETS data (Kong et al. 2023). Other attempts have relied on the use of online directories such as Yelp, YellowPages, and GuidetoVaping.com to catalog locations nationally (Dai et al. 2017; Venugopal et al. 2020), in addition to Google (Kim et al. 2016).

This study builds on previous research on specialty tobacco store location in several ways. First, nationally representative database of tobacco specialty stores indexed by Google with consumer visit counts available in Advan. Our registry is particularly useful as these stores types are rarely included in retail scanner data sources, thus little is known about them. Our data, collected in June 2024, provides the first national estimates since (Venugopal et al. 2020), who provided estimates for just vape shops but not other specialty store types such as cigar, hookah, and tobacco like we do in this paper. Second, we distinguish between different specialty tobacco retailers. Distinguishing between these store types is crucial as store types are different in their product offerings and marketing strategies. Third, as a proxy for access, we link store locations to foot traffic data as recorded by Advan, which uses cell phone tracking data across over 40 million devices. We find that additional stores generally increase visitor counts in the census tract, rather than redistributing existing customers. Finally, we

tag tobacco specialty retailer locations to the census tract level. Census tracts are small entities within counties (or the statistical equivalents of counties) delineated by a committee of local data users. Tracts are defined to be as homogeneous as possible with respect to population characteristics, economic status, and living conditions (US Census Bureau). This level of geographic detail is important because there is often considerable variance of economic characteristics within common geographic boundaries such as metropolitan areas or counties.

Data and Methods

We accessed Advan monthly foot traffic files via a subscription to DeweyData.io ([Advan Research 2025](#)). Advan aggregates Safegraph’s cell phone mobility data, which tracks user movement to over 150 million points of interest to report foot traffic counts. At the time of analysis, Advan data included approximately 40 million cell phone users, representing about 13% of the number of cell phone users in the U.S. Consumers who enter a geographically defined polygon with an app active which transmits location data are included in the Advan data.

We create a cross-section of tobacco specialty stores in May 2024 by combining Google Maps data with Advan’s monthly foot traffic point of interest data. The main benefits of Advan’s data are that it is accompanied by monthly foot traffic counts generated from consumer cell phone mobility data. However, the Advan data does not record the type of tobacco specialty store, and in some cases wrongly attributes a store that was previously a tobacco specialty store as remaining a tobacco specialty store even though the store has closed and reopened as a different type of business. The Google data is used to verify that the store remains open and a tobacco specialty store.

To facilitate the identification of tobacco specialty stores, we started with all Advan data for a particular North American Industry Classification System code, a six-digit number used to classify businesses by industry. NAICS code (453991 refers to tobacco, electronic cigarette, and other smoking supplies retailers, and contained all instances of vape, smoke and cigar shops in the Advan dataset. While Advan provides NAICS code categorizations and brief categorical store type descriptions in its dataset, it does not contain any identifier for specific store types within a NAICS code. Within the Advan database, we find 43,640 unique stores listed under NAICS code 453991 in May 2024. Among those, we retain only the subset of 29,642 locations with positive foot traffic counts.

An important shortcoming of the Advan data is that points of interest in the database are occasionally added, but never removed and the database does not consistently tag closures. As a result, it is possible for a store to remain in the Advan database generating foot traffic counts, while in reality having closed and converted to a different store, or in some cases a new store type. Matching to Google data acts as a secondary verification method for the set of open stores,

in addition to providing information about the types of products sold at the store.

Hookah stores were not contained within the same NAICS code and required a search across multiple codes including restaurants and bars, in addition to other tobacco retailers. We first created a list of all NAICS codes which contained stores with the phrase ‘hookah’ in the store name. We extracted all observations contained in this list of NAICS codes and used that list as our source for potential matches. Because hookah was not included in NAICS code 453991 in the Advan database, the list of potential matches for hookah retailers is different and unique from the list used for vape, smoke and cigar shops.

To create a database of stores publicly visible in Google Maps, we used a third party program (outscraper.com) to iteratively query Google Maps in June 2024. By ZIP code, we searched for the following list of store types: vape store, tobacco shop, smoke shop, cigar shop, hookah bar, and hookah store. Queries were submitted by individual store type and zip code, for example “Vape Stores in Missouri, 65201”. This resulted in unique queries for each of the six store types in each of the 41,704 ZIP codes in the United States (250,224 distinct queries = 6 store types * 41,704 ZIP codes). We found 55,333 unique stores with reviews and full address detail; of these, 16,416 identified primarily as vape shops, 32,740 tobacco shops, 3,365 cigar shop, and 2,812 hookah shops. We found that it was possible for businesses to appear in multiple queries, based on secondary business type, reviews, or webpage keywords. We removed any duplicates by retaining only the open location at each address with the largest number of reviews. Finally, using the main store type categorizations visible on the Google Maps page, we filtered stores into main categories: vape shops, smoke shops, cigar shops and hookah shops, based on their primary type listing in the Google database. Additional information about Google’s store type categorizations can be found in our supplemental online appendix.

There are about 12,000 more specialty tobacco stores listed in Google than in Advan. This is possible for a number of reasons. First, Advan does not continuously track openings and closures of store locations and has not updated its database since about 2023. Any openings between 2023-2024 would not be included. Second, the Google listings may include some erroneous, duplicate, or nonexistent locations which we removed from the database. We combined the store data from Google with Advan data by using geographic coordinates, business names, and addresses in each database, and applying a fuzzy matching process. Additional information about our matching method is available in the online appendix. By matching the Advan database with the Google database, we identified 22,484 specialty locations actively operating in the United States, including 6,295 vape stores, 13,021 tobacco shops, 1,507 cigar shops, and 1,661 hookah shops. This combined dataset provides comprehensive insights into foot traffic at tobacco specialty retailers nationwide. For May 2024, Advan reported nearly 6.5 million visits across the 22,484 cross verified tobacco specialty stores,

averaging 289 visits per location. This number, if scaled to account for all cell phone users and non-cell phone visitors, would result in approximately 50 million total visits to these stores. Among the visits in our sample, 28% were to vape shops, 54% to tobacco shops, 11% to cigar shops, and 7% to hookah shops.

We use the finalized May 2024 cross-section to estimate the association between census-tract level tobacco specialty store measures and foot traffic. We conduct this analysis at the census tract level, controlling for census-tract covariates such as population percent ages 15-24 (NIH), minority percent (NIH), female percent (NIH), poverty percent (ACS), percent with high school degree or less (ACS), and an urban/rural indicator (USDA). In our regression analysis, we also use weights of the census tract’s population size.

Results

To illustrate the geographic dispersion of store types in the US, we aggregate store counts to the US county and census tract levels. Figure 1 displays the density deciles of all combined vape, tobacco specialty, cigar and hookah stores, scaled per 100k residents in each county. Density appears to align with expectations in a few key places. For example, in Figure 1, Clark County, Nevada (home of Las Vegas) and Appalachia have high density of tobacco specialty stores; meanwhile, Utah (home to a sizable Mormon population) has low density. Figure 2 meanwhile shows each store type separately across four maps. Access to electronic cigarettes in terms of retail availability may be strongly influenced by local legislation, such as San Francisco’s ban on e-cigarette sales in 2019 (Yang et al. 2020; Friedman 2021). Cigar and hookah store locations are less common nationally than vape shops and tobacco specialty stores. Cigar shops appear disproportionately present in Pennsylvania, Florida, and Washington DC that are the only states / district in the USA that does not tax large cigars (Centers for Disease Control and Prevention, Office on Smoking and Health, National Center for Chronic Disease Prevention and Health Promotion 2024).

We include 73,245 census tracts across the United States during May 2024. Supplementary online appendix Figure 6 presents summary statistics for the tracts included in our study. The average census tract had 0.09 vape (sd = 0.33), 0.18 tobacco/smoke shops (sd = 0.49), 0.02 cigar shops (sd = 0.16), and 0.02 hookah shops (sd = 0.18). Tracts with stores contain an average of 1.13 vape shops (sd = 0.42), 1.22 tobacco/smoke shops (sd = 0.55), 1.05 cigar shops (sd = 0.25) and 1.15 hookah shops (sd = 0.53). The average census tract has 13.13% (sd = 8.26) of its population between the age of 15-24, 38.69% (sd = 29.96) of its population identified as categories other than White non-Hispanic, the average poverty rate was 14.65% (sd = 11.59), and 69.29% (sd = 19.32) without a bachelor’s degree. Census tracts are either all urban and rural, and

37.45% are rural.

Supplementary appendix Figure 7 presents the percentage of census tracts which contain 0, 1, 2, 3 and 4+ stores, along with the average number of visitors in those tracts. It is interesting to note that while smoke shops are the most common store type, they are still absent from over 85% of all census tracts. 92%, 98% and 98% of tracts do not have vape shops, cigar shops and hookah shops, respectively. Among vape and smoke shops, visits scale approximately linearly. For example, in tracts with one vape (tobacco) shop, there are an average of 207 (182) visitors, while in tracts with two shops there are 425 (396) visitors and in tracts with three shops there are 644 (621) visitors. There are very few census tracts which contain more than two of more of any store type, but multiple cigar and hookah shops within a tract are especially rare.

Figure 3 presents regression coefficients from an ordinary least squares regression at the census tract level with state fixed effects and Huber-White robust standard errors. The most salient of our findings is a large positive relationship between poverty and density of all store types. In particular, a 1% increase in the tract poverty rate is associated with 0.02 extra vape shops, 0.08 extra tobacco specialty stores, 0.01 extra cigar stores, and .03 hookah stores (all per 100,000 people, $p < 0.01$).

A second consistent finding across all store types is that rural areas have less tobacco specialty stores per capita. Tracts in rural counties, as designated by the USDA Economic Research Service (ERS) rural urban continuum codes, had on average 0.011 fewer vape shops, 0.013 fewer tobacco/smoke shops, 0.002 fewer cigar stores, and .003 fewer hookah stores (all per 100,000 people, $p < 0.01$). This may or may not be considered surprising as smoking rates are generally higher in rural areas (Doogan, et al., 2017) but specialty shops are less likely to operate (Reed, et al., 2023), which we highlight in the discussion.

In addition to the relationship with poverty and rurality, we find a generally negative relationship between tract minority share, as defined by total population minus white non-Hispanic, and vape and smoke shop density and visits. A one percent increase in the density of minority residents corresponds with 1.5% fewer vape shops and three fewer visits to those locations, as well as 1.1% fewer tobacco shops and 2.5 fewer visits. While previous literature has found higher vape shop density associated with Black and Hispanic populations (Dai, et al., 2017), that association was weakening over time 2000 to 2017 (Mills, et al., 2022). Our results provide additional evidence that the landscape of specialty tobacco retailers may be changing, especially in the post-pandemic era.

We find that population share ages 15-24 is positively and significantly related to both vape store density and foot traffic visits to vape stores. A one percent increase in the population density between ages 15-24 corresponded to an increase of about .01 locations per 100k population, as well as an increase of about

6 visits to vape specialty stores in each tract.

By identifying specific store types, we provide an estimate both of the extensive margin, the impact of having any specialty store in a tract, and the intensive margin, the impact of one additional store of any type. Access is measured by the number of distinct visitors at each store in each period. In Figures 4 and 5, visitor counts are logged so that coefficients are interpreted as semi-elasticities measuring the impact of an additional store on the percentage change in foot traffic to stores of that type in the tract.

Figure 4 presents measures of the impact of an additional store of each type on visitor counts first in all tracts in column 1, and only in tracts with stores of the type in column 2. We find that an additional vape shop increases foot traffic to vape shops by about 366%, driven largely by the impact of the first store, with similar effects for an additional smoke shop (311%) and an additional cigar shop (420%). We find a smaller magnitude for the effect of an additional hookah shop (56%) (all $p < .01$). Including only tracts with stores, we find smaller effect sizes, suggesting that these results may be largely driven by the impact on access of the first store location in the tract. In tracts with positive store counts, an additional vape shop location is associated with an 87% increase in visitors to vape shops in the tract, with similar effects for smoke shops (85%), cigar shops (117%), and hookah shops (46%, all $p < .01$).

While effect sizes are largest within store, we also find that additional store locations of each type are associated with statistically significant increases in visitor counts at other types of stores in the tract. These results are listed in Figure 5. For example, an additional vape store location was associated with increased food traffic to smoke shops (56%), cigar shops (79%) and hookah shops (70%) (all $p < .01$).

This result suggests that there may be some complementarity in access between the different store types. This is possible since there may be considerable overlap in product offerings between different specialty tobacco stores, despite the focus and expertise being in a particular product or product group. Vape shops and smoke shops, which often have more diverse product offerings, have larger effects on foot traffic at other tobacco specialty store types than cigar and hookah shops. Still, these cross effects are consistently positive and significant, suggesting that additional tobacco specialty stores of any type may be associated with higher foot traffic at other store types.

Discussion

This study provides a national assessment of specialty tobacco retailer locations and consumer foot traffic, highlighting disparities in tract level measures such as urban/rural, poverty rates, youth exposure, and racial composition. Our results provide evidence confirming prior research showing higher concentrations of

tobacco retailers in low-income communities (Rodriguez et al. 2013; Mills et al. 2022), but with the potentially surprising caveat that these specialty retailers are also less likely to operate in rural areas where smoking prevalence is often higher (Doogan et al. 2017). After adjusting for population by scaling stores per 100k population and population weighting our coefficient estimates, we find that all stores are more likely to locate in urban than rural tracts. These results are also unsurprising given that other store types such as specialty grocers (Reed et al. 2023) are also much more likely to operate in urban areas. This suggests that while there are more saturation in urban markets, specialty tobacco shops are still choosing to locate there. In many cases these shops provide access to employees who are more knowledgeable about the products than employees of large retailers.

This pattern raises important public health concerns related to tobacco access, as economically disadvantaged populations already experience higher rates of tobacco use (Farley et al. 2019; Kingsbury et al. 2020), increased exposure to environmental stressors (Hobkirk et al. 2018), and greater barriers to smoking cessation resources (Wijk et al. 2019). Tobacco access in low-income areas has been linked to persistent disparities in smoking prevalence, as greater exposure to tobacco outlets is associated with higher smoking initiation rates among youth and reduced cessation success among adults (Baker et al. 2023). To date, our study is the first to consider this relationship at the census tract level. We find that a one percentage point increase in poverty is associated with a statistically significant increase in both the density of tobacco and vape specialty stores per capita and the total foot traffic visits to these stores.

Poverty is strongly associated with both store density and consumer foot traffic, raising concerns about targeted marketing and persistent health disparities. Given that increased tobacco retailer access is linked to higher smoking initiation rates and lower cessation rates (Farley et al. 2019; Reed et al. 2023), policies aimed at regulating tobacco retail density in disadvantaged areas may help reduce disparities in smoking-related health outcomes. Cigar and hookah stores exhibit a positive but weaker relationship with poverty, which may reflect different target consumer bases and purchasing behaviors. While past studies have found higher tobacco retailer density in Black and Hispanic communities (Dai et al. 2017; Mills et al. 2022), our results suggest a shifting pattern where poverty, rather than racial composition, is a significant predictor of retailer density. Some of the most stringent tobacco control measures have been passed in urban areas with high minority densities such as San Francisco, New York and Chicago.

The positive association between vape store density and youth population (ages 15–24) previous research highlighting spatial clustering of stores near schools (Halvorson-Fried et al. 2024). This result underscores concerns about youth access to nicotine products despite sales restrictions which may exist. However, we do not find a similar relationship for tobacco, cigar, or hookah stores, sug-

gesting that youth exposure concerns are more specific to vaping products as compared to “dirtier” alternatives. Retail environment may be closely associated with prevalence of use, with potentially detrimental impacts on populations with high exposure ([Glasser and Roberts 2021](#); [Glasser et al. 2022](#)). As debates continue over flavor bans and zoning restrictions, future research should examine how policies affect both youth exposure and adult access to harm reduction products.

This study is subject to several limitations. First, we only use a single cross section of data. Historical Google data required to distinguish between store types each month of the Advan data panel was not available. Future work will include efforts to establish a running panel of store locations from changes over time in the Google database.

We do not produce a complete and exhaustive list of all operating tobacco specialty stores in the United States. Our set contains only the matches between the Advan and Google databases. Additional information on our matching algorithm can be found in the online data appendix. Nonetheless, our method produces the most up to date overview of specialty tobacco retailers possible while being reproducible and updatable to future data. No other published work to date includes nationally representative data on specialty tobacco retailer locations in 2024 or more recently.

There are other store types including, but not limited to: grocery stores and supermarkets, corner stores, gas stations, and convenience stores which may also sell the same products found in specialty tobacco stores. A large portion of total sales do occur at these types of retail outlets. We do not include these store types, due in part to complications in determining in a national database whether an individual store location has a tobacco retail license. Furthermore, our classification of store types may be subject to inaccuracies within the Google Maps database. Online registries with a single business category defined by the business or by other methods may not accurately capture the product offerings sold at the store location. For example, vape shop locations drawn from Yelp had a high probability of selling tobacco products in addition to e-cigarettes ([Giovenco 2018](#)).

This is especially important when considering vaping products as smoking cessation aids. Instead, we consider a more general definition of access to specialty tobacco products, which encompasses but is not limited to cessation aids. The past few years have been an active time period for policy, including adoption of Tobacco-21 nationally across many states and nationally ([Abouk et al. 2024](#)) ([Abouk, et al., 2024](#)), tobacco product flavor restrictions ([Friedman et al. 2023, 2024](#)), and the FDA’s review of millions of pre-market tobacco product applications (FDA Tobacco Products Marketing Orders). While we do not directly consider the effects of these policies on store location or foot traffic in this work, future work will consider how regulation can impact tobacco access as measured

by store location and volume of foot traffic.

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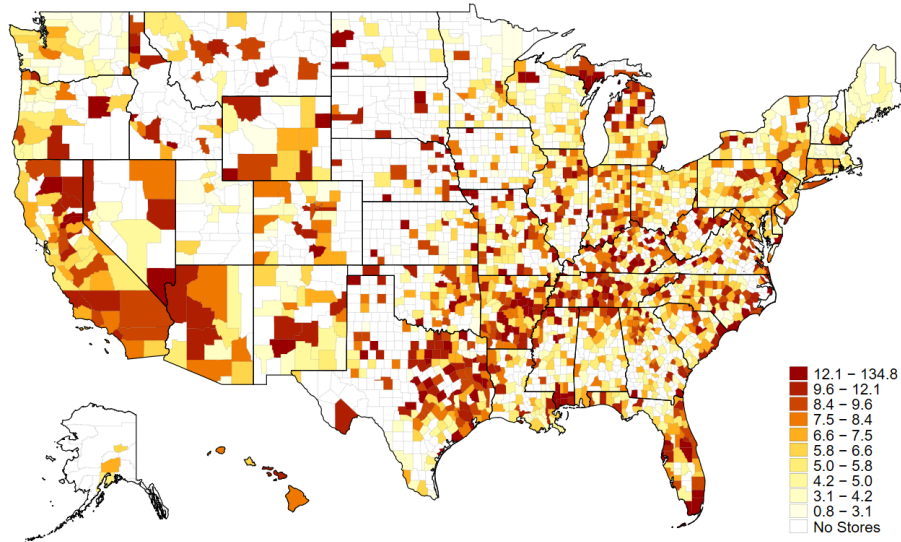
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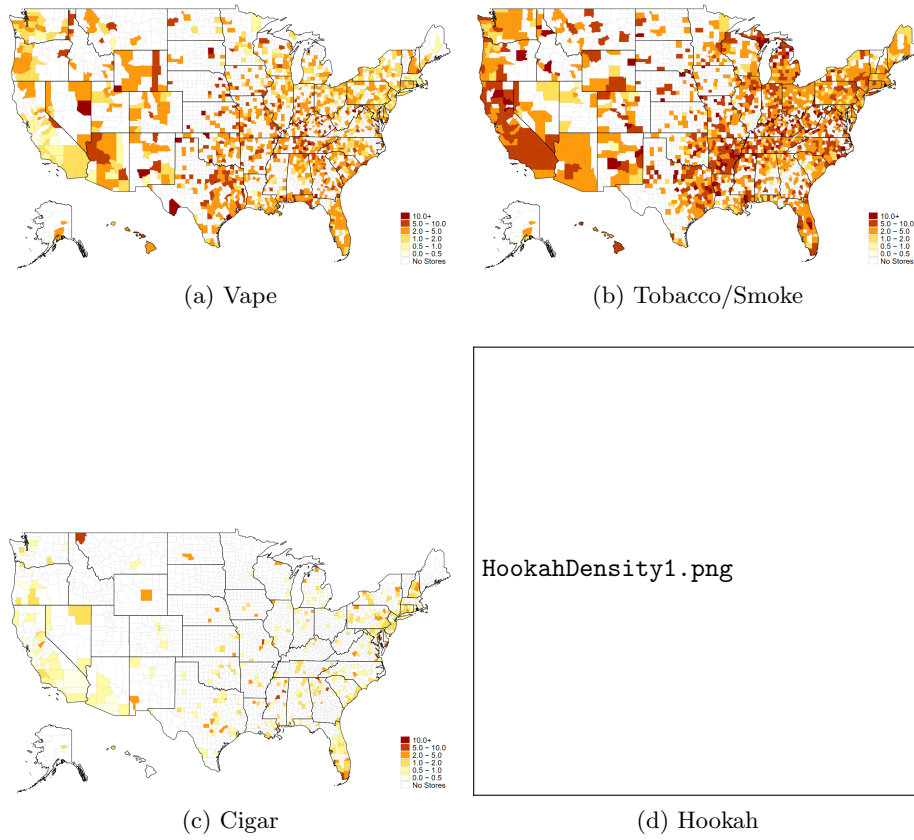
Results Figures

Figure 1: County Store Density, per 100k Population



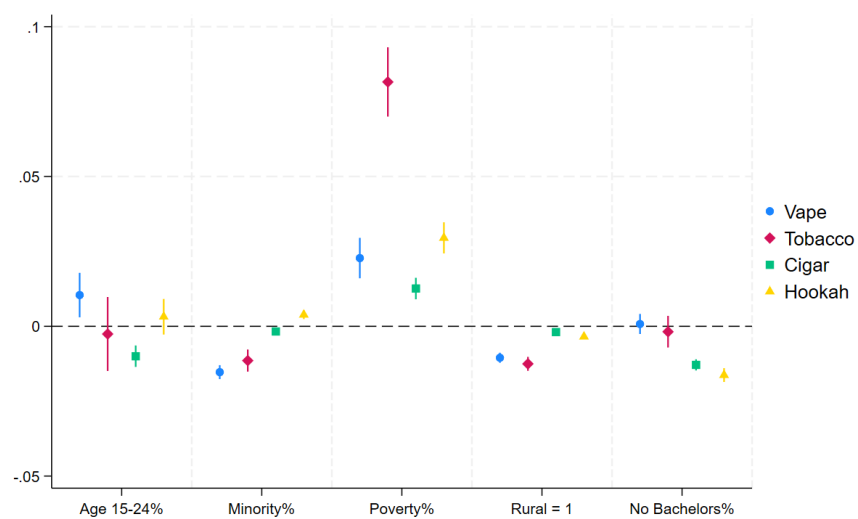
Note: Figure 1 displays deciles of the number of combined vape, tobacco/smoke, cigar, and hookah specialty stores per 100k county residents, based on matches between Advan and Google data.

Figure 2: Density of Store Types by County, per 100k Population



Note: Figure 2 displays independently deciles of the number of vape, tobacco/smoke, cigar, and hookah specialty stores per 100k county residents, based on matches between Advan and Google data.

Figure 3: The Association of Census Tract Demographics with Store Type Density



Note: Coefficients and standard errors are estimated from OLS Census Tract level regressions with population weighed, robust standard errors and state fixed effects. Intervals are drawn to show 95% confidence bands.

Figure 4: Foot Traffic on Store Density Coefficient Regression Coefficients

	All Tracts ln(Visitors)	Tracts with Shops ln(Visitors)
# Vape Shops	3.630*** (0.057) N = 72410	0.870*** (0.061) N = 5828
# Smoke Shops	3.114*** (0.045) N = 72410	0.852*** (0.037) N = 10945
# Cigar Shops	4.202*** (0.064) N = 72410	1.171*** (0.250) N = 1571
# Hookah Shops	0.562*** (0.044) N = 72410	0.458*** (0.126) N = 1520

Robust standard errors in parentheses. Fixed Effects: State

* $p_{i.1}$, ** $p_{i.05}$, *** $p_{i.01}$

Note: Each cell represents a unique regression of the store type in the left column on visitors to stores of that type in the tract. The columns represent different specifications: 1) a binary indicator for whether any store of the type exists in the tract, 2) regression on only tracts with stores of the type, and 3) regression using all tracts. Coefficients and standard errors are estimated from OLS census tract level regressions with population weighted, robust standard errors and state fixed effects.

Figure 5: Foot Traffic on other Store Type Densities

	All Tracts ln(Visitors)			
	ln(Vape Visitors)	ln(Smoke Visitors)	ln(Cigar Visitors)	ln(Hookah Visitors)
# Vape Shops	3.630*** (0.057)	0.516*** (0.022)	0.785*** (0.065)	0.702*** (0.062)
# Smoke Shops	0.093*** (0.036)	3.114*** (0.045)	1.176*** (0.073)	1.236*** (0.075)
# Cigar Shops	0.192*** (0.018)	0.156*** (0.013)	4.202*** (0.064)	0.304*** (0.042)
# Hookah Shops	0.050*** (0.011)	0.040*** (0.006)	0.068*** (0.018)	0.562*** (0.044)

Robust standard errors in parentheses. Fixed Effects: State

* $p_{i,1}$, ** $p_{i,05}$, *** $p_{i,01}$

Note: Each cell represents a unique regression of the store type in the left column on visitors to each store type in the tract. Entries on the diagonal are the results from the third column of Figure 4, including all census tracts. Coefficients and standard errors are estimated from OLS census tract level regressions with population weighted, robust standard errors and state fixed effects.

Online Data Appendix

Advan Data

We used a subscription to DeweyData.io to access Advan monthly foot traffic files. May 2024 was the most recently available month of data at the time of download. The Advan database includes store name, address, and geographic coordinates in addition to NAICS code, description and monthly foot traffic counts for over 800 million points of interest (POI) worldwide. We focused on NAICS code “453991”, which housed all instances of store types including the word ‘vaporizer’ or ‘vape’. While NAICS categorizations for vape and tobacco products have changed over time, Advan keeps the definition constant throughout the database through the months included in our dataset.

There are a few minor but important limitations of the Advan data. First, because POI are never removed from the database, it is not always apparent which locations are operational in May 2024, and which are still in the database but have closed and been replaced. While prior updates to the database did retroactively mark POI as closed from the previous months’ data, closures are not consistently updated within the database and so the database does not accurately and completely specify which locations remain open. Second, within the NAICS code there is no way to distinguish between store types. The 453991 code includes all types of stores which sell tobacco products. Since regulations may impact only certain stores or store types, it is important to be able to distinguish between the different store types within the NAICS code.

Google Data

Google Maps displays publicly available data including store name, category, and address. We queried the Google Maps database to match store type information to the addresses included in our the Advan database. For each ZIP code in the United States, we created a query for each of six store types: vaporizer store, tobacco shop, smoke shop, cigar shop, hookah bar, and hookah store. While Google continuously updates its database to remove fictitious or erroneous entries, we remove any locations with zero reviews, along with any entries with ‘temporarily’ or ‘permanently’ closed designations to ensure that we are identifying only open and operational store locations. By this process, we identify over 60,000 possible locations of these store types in the Google database. After removing address duplicates, we employ a fuzzy matching process based on geographic coordinates, store name and address.

Since hookah is also a tobacco product sold in retail outlets, we expand our NAICS code definition to capture retail locations selling hookah products. By inspection of the Advan database, we identify all NAICS codes containing a business with the phrase “hookah” in the name, and retail all entries in that set of NAICS codes. Using an identical matching algorithm across the expanded

Advan NAICS target list, we identify an additional 1,755 matches for hookah retailers in the United States. The cross referencing of these locations across the two databases increases our confidence in identifying open locations at a large geographic scale.

Google collects store information using two methods. Owners can register a Google business profile and select a store type among the list of options. For listings without an active business profile, Google crawls the web to analyze content potentially related to the store location, including user-generated content, website information and online reviews.

Matching Advan to Google Locations

We matched Advan listings to Google database locations using a fuzzy matching algorithm in STATA on business name, address, and geographic coordinates. From Advan, the data to be matched for vape, smoke and cigar shops included all observations from NAICS code ‘453991’. Hookah stores were dispersed across a variety of NAICS codes and so the potential matches for hookah stores included all observations in Advan across any NAICS code which included any store with the phrase “hookah in the name”. From these, we retained only observations in Advan with positive foot traffic counts in May 2024. Using the Google main business categorization, we identify 6,597 vape store matches, 13,397 specialty tobacco store matches, and 1,646 cigar store matches.

While each database contains latitude and longitude coordinates, store name and address, there are slight differences across the database which required the implementation of a fuzzy matching algorithm. The steps of the algorithm and their rationale is detailed as follows:

1. For each point of interest in Advan, use ‘geonear’ in STATA to produce a list of all vape, smoke, cigar and hookah shops in the Google database within a 1 kilometer radius.
2. Among each potential match in the 1 kilometer radius, use ‘matchit’ in STATA to produce a name match score and a location match score based on full street address and business name strings, converted to lowercase. Define a match to be name likeness score and address likeness score of over .5, with tagged distance within 100 meters. This step produces a rough matching criterion which needs to be further refined.
3. For each potential match, use ‘matchit’ in STATA to produce a name and address similarity score based on only the first six characters of the strings. Define a match to be name or address likeness score of over .8 within 100 meter radius, or an exact match of 6 character name and address score within 1 kilometer. The previous step fails to find matches if the names, addresses, or coordinates are too different. Occasionally the name or address in one or the other database contain extra words or

characters leading to match inaccuracy.

4. Identify near exact name and address matches outside the 100 meter radius but still within 1 kilometer. There are some discrepancies in coordinate geotagging between the two datasets for clear name and address matches. These cases are identified in this step.
5. Exact address matches are counted as matches. For example, if we know there is a store in the correct NAICS code in the Advan database, and there is a vape shop with a different name at the same address, we count this as a match. While future research will seek to identify store closures and ownership changes, we are only interested in the existence of and foot traffic to a location.
6. Remove newly tagged matches with full name likeness score less than .5 and distance greater than 500 meters, or with six character name and address likeness scores each under .6. Previous steps geared toward including matches also identified some false positives which need to be eliminated. This step filters back out pairs which had very similar names and addresses, but turned out not to be close enough to be considered a match.
7. Use 'split' command to extract and identify the numeric component of a store address. Identify any untagged matches with exactly identical numerical address within a 50 meter radius. This step catches a few final matches which had some entry errors or differences in character strings, but were clear matches given their store type and proximity.

Advan does not delete points of interest from their panel, and only occasionally adds from their original aggregates in January 2019, and while there is an indicator for business closure, this data is not routinely updated. We consider entries in Advan not found in Google as likely to have closed between the beginning of 2019 and May 2024 and remove them from our dataset.

There are also entries in Google which are not included in Advan. There are a variety of potential reasons including incorrect store type tagging in Google, user added stores in the database which don't exist but haven't been flagged by their web crawling algorithm, and permanently closed locations which occasionally appeared in queries of rural areas where there were few open matches. Since we could not verify the authenticity of these entries and did not have foot traffic data at these locations to use as a volume capacity proxy measure, we also deleted Google stores not found in Advan. While we do not include extensive study of the stores from each database in this manuscript, we did verify that there was no evident geospatial pattern in the points of interest omitted from each database.

Appendix Figures

Figure 6: Tract Summary Statistics

	mean	sd	min	max
Tract Vape Shops	0.0901	0.3282	0	11
Tract Tobacco Shops	0.1829	0.4850	0	12
Tract Cigar Shops	0.0225	0.1563	0	5
Tract Hookah Shops	0.0240	0.1811	0	10
Age 15-24%	13.1354	8.264425	0	100
Minority%	38.6924	29.9588	0	100
Female%	50.7418	4.7931	0	100
Poverty%	14.6480	11.5919	0	100
Rural	16.8667	37.4460	0	100
No Bachelors%	69.2966	19.3208	0	100
Vape Shop Visitors	18.8164	169.0991	0	15608
Smoke Shop Visitors	34.6084	187.7917	0	15236
Cigar Shop Visitors	6.6909	266.3809	0	60205
Hookah Shop Visitors	0.5870	20.4129	0	3783
Observations	73245			

Note: Figure 6 presents summary statistics for all variables included in the analysis presented in Figures 3-5.

Figure 7: Frequency of Tract Store Density and # Visitors

	# of Stores per Tract				
	0	1	2	3	4+
Vape Shops	92.2%	7.0%	0.8%	0.1%	0.1%
Avg. Visitors	-	207	425	644	1346
Smoke Shops	85.3%	12.1%	2.1%	0.4%	0.1%
Avg. Visitors	-	182	396	621	833
Cigar Shops	97.9%	2.0%	0.1%	0.0%	0.001%
Avg. Visitors	-	261	1405	731	2913
Hookah Shops	98.0%	1.8%	0.2%	0.0%	0.0%
Avg. Visitors	-	25	35	52	239
Total Tracts	74,326				

Note: Figure 7 presents the percentage of tracts which contain 0, 1, 2, 3, and 4+ stores of each type. For each type of store and store count, we display the average number of visitors to stores in the tract. For example, in tracts with one vape shop, Advan records an average of 207 monthly visitors the location.